

Highly Scalable and Reliable Travel Booking Website

System Design

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One Pager Summary

Goal: Deliver a global travel marketplace for searching accommodations and making reservations with accurate availability, competitive pricing, and secure payments.

Users: Guests, hosts and hotels, channel managers and suppliers, operations and support, finance.

Core Value: Fast search, trustworthy availability and pricing, frictionless checkout, and resilient confirmation flows with supplier integrations.

North Star Metrics: p95 search latency under 300 milliseconds for cached routes, booking confirmation success rate above 99.5 percent, payment auth success above 96 percent, availability accuracy above 99.9 percent, platform availability 99.95 percent.

Scope: Inventory ingestion, search and pricing, availability and rate checks, reservation workflow, payments and refunds, cancellations and modifications, messaging, analytics, and compliance.

Functional Requirements

Search: location or map based search with dates, guests, filters (price, amenities, rating), sorting (price, distance, popular).

Availability and pricing: real time or near real time rates and restrictions (min nights, blackout, refundable).

Currency conversion.

Listing detail: photos, amenities, policies, fees and taxes breakdown, reviews, host info, cancellation terms.

Reservation: hold inventory, collect guest details, apply promos, calculate totals, pay, confirm, email or push confirmations. Support modify and cancel with policy enforcement and proration.

Accounts and identity: sign up, login, MFA, KYC for hosts, profile, saved payments, saved stays, wishlists, recent searches.

Supply: host portal for calendar, rates, bulk upload, and channel manager or PMS and GDS integrations.

Ops: manual overrides, refunds, vouchers, case workflows, fraud review, finance exports.

Internationalization: locales, currencies, time zones, tax rules, PSD2 SCA where required.

Accessibility: WCAG compliant, screen reader and keyboard support.

Observability: tenant and supplier level dashboards for latency, error, and conversion.

Non

Availability 99.95 percent platform wide. Read endpoints (search) 99.99 percent from edge cache.

Latency targets: p95 search under 300 milliseconds for cache hits and under 800 milliseconds for cache misses with supplier round trips; p95 booking creation under 3 seconds end to end; p95 payment auth under 1 second.

Consistency: Read your writes for bookings; eventual consistency for search index and analytics within minutes. Strong consistency for inventory holds and confirmations within a reservation saga.

Durability: six nines for operational data and eleven nines for payment and booking event logs.

Scalability: tens of millions of daily searches, hundreds of thousands of daily bookings, bursty traffic during holidays and events.

Privacy and compliance: SOC 2 Type 2, PCI DSS for payments, GDPR and CCPA with data minimization, retention, and subject request tooling.

Cost efficiency: multi tier caching for search, async supplier polling and prefetch, tiered storage for logs and images.

Architecture Overview

Client: Web and mobile apps using a shared TypeScript SDK. Server side rendering for page shell and streaming for results, client hydration for interactivity. Cached search results and prefetch of details. Feature

flags for experiments.

Edge: CDN and edge compute for HTML and JSON caching, geolocation, token checks, bot mitigation, and A/B assignment. Edge KV for hot searches and autosuggest.

Gateway: API Gateway with WAF and DDoS protection. REST and GraphQL. Request routing by tenant and region.

Core Services

Identity Service: auth, MFA, session, OAuth for partners.

Search and Discovery Service: reads from listings index, applies filters, facets, sorting, and personalization.

Pricing and Availability Service: evaluates rates, restrictions, currency, and taxes; calls suppliers when needed.

Reservation Service: orchestrates booking saga, holds inventory, confirms or releases, issues documents.

Payment Service: tokenized payments, PSD2 SCA, capture, refund, ledger, payouts.

Inventory

Ingestion

Service: pulls or receives pushes from PMS, channel managers, and GDS; normalizes ARI (availability, rates, inventory).

Listing Service: canonical listing data, photos, amenities, policies.

Review Service: ratings and reviews with anti abuse.

Notification Service: email, SMS, and push.

Partner Integration Service: connectors to PMS, channel managers, GDS, and tax engines.

Fraud and Risk Service: device and behavioral signals, rules, and ML.

Analytics and ETL: event collection and warehouse.

Admin and Ops: tooling and case workflows.

Control Plane: configuration, feature flags, autoscaling, and deployments.

Data Flow

Search flow

Client sends a debounced search with location, dates, and filters. Edge cache serves popular queries. On miss, Search Service queries the index for eligible listings, joins with precomputed price buckets, and materializes a result page. For uncertain dates or dynamic suppliers, a parallel fresh availability check refines displayed prices with inline updates.

Reservation flow (saga)

1 create intent and price quote, 2 place hold on inventory with expiry, 3 authorize payment, 4 confirm with supplier and commit booking, 5 capture payment, 6 issue confirmation and receipts. On failure, compensating actions release holds, void auths, and notify user.

Supply flow

Hosts update calendars and rates in portal or via PMS.

normalizes ARI updates to internal format,
applies validation, and publishes to index and cache with versioning and audit.

Modification or cancellation

Reservation Service computes penalties per policy and triggers refunds or additional capture via Payment Service. Supplier adapters propagate changes and confirmations.

Technology Stack

Client: TypeScript, React or Next, React Query, Service Workers for offline receipts and retry.

Edge: CDN and edge compute with KV for hot queries, image resizing.

Gateway: Envoy or API Gateway and WAF.

Search index: OpenSearch or Elastic for geo queries, faceting, and relevance. Vector search for personalization.

Operational stores: Postgres for bookings, listings, and finance with read replicas and row level security.

Redis for caches, locks, and rate limits. DynamoDB or Cassandra optional for write heavy ARI streams.

Messaging: Kafka or Kinesis for ARI updates, booking events, and analytics.

Payments: PSP integrations, tokenization vault, 3DS for SCA, webhooks ingestion.

Geo and tax: geocoding and tax engine integrations.

Observability: OpenTelemetry, Prometheus, Grafana, centralized logs. RUM for client performance.
Infra: Kubernetes, Terraform, canary deploys, and feature flags. Object storage for images and documents.
Data Model Sketch

Tenant: id, name, settings, region, plan, keys, status, createdAt.

User: id, tenantId, email, phone, authProvider, kycStatus, createdAt.

Listing: id, tenantId, hostId, title, description, location lat lon, address, amenities, photos, policy, rating, createdAt, updatedAt.

RatePlan: id, listingId, currency, nightly price rules, fees, taxes, minNights, refundPolicy, blackout rules.

Availability: listingId, date, allotment, closed flag, version, updatedAt.

PriceQuote: id, listingId, checkIn, checkOut, guests, priceBreakdown, currency, expiresAt, version.

Booking: id, tenantId, listingId, userId, checkIn, checkOut, guests, status, total, currency, policySnapshot, supplierRef, confirmationCode, createdAt.

Payment: id, bookingId, method, provider, amount, currency, status, authId, captureId, refundIds, createdAt.

Hold: id, listingId, date range, units, expiresAt, status, correlationId.

Event: id, type, bookingId optional, payload, ts.

Availability Pricing and Taxes

Availability

Single source of truth per listing calendar with versioned updates. Use compare and set when applying ARI to avoid lost updates. Holds decrement provisional inventory until expiration or commit.

Pricing

Price engine composes nightly rates, fees, and taxes using rule graph and currency conversion. Return price breakdown to the client with total and cancellation penalty schedule. Cached price buckets by date ranges support fast search.

Taxes and fees

Integrate with regional tax engines and fee calculators. Maintain jurisdiction rules and thresholds. Apply inclusive or exclusive tax pricing by locale and show itemized totals.

Algorithms and Workflows

Search ranking blends price, distance, quality, popularity, and personalization signals using a learning to rank model. Diversify results across neighborhoods and property types to avoid crowding. Geo index uses geohash or H3 for map queries.

Reservation saga uses outbox pattern and idempotency keys per step. Payment uses auth then capture, with 3DS challenge handling. Retries are safe through idempotency and time bounded holds. Dead letter queues capture failed steps for ops.

Fraud scoring uses device fingerprint, IP reputation, velocity, chargeback history, and graph signals; risky bookings require SCA or manual review.

Currency conversion uses mid market rates with a markup and daily caching. For high value bookings, real time rates are fetched.

Storage and Caching Strategy

Search cache layers: edge KV for top routes and weekend ranges, Redis for regional caches, index for long tail. Price buckets cached with keys by listing, date range, and occupancy. Calendar cache with fine grained invalidation on ARI updates.

Bookings and payments in Postgres with partitioning by month and tenant. Hot rows in Redis with write through on confirmation. S3 object storage for images, invoices, and policy PDFs with CDN for delivery.

Event logs in object storage and data lake for analytics. Data retention and deletion policies per region and tenant.

Latency Budgets and Throughput

Search p95: 300 milliseconds cache hit, 800 milliseconds cache miss. Booking create p95: 3 seconds including supplier call and payment auth. Payment auth p95: 1 second, capture p95: 800 milliseconds. ARI ingestion supports 50 thousand messages per second per region. System handles 1 million searches per minute and 50 thousand bookings per hour globally with auto scaling on queue depth and CPU.

SLA and SLOs

Platform availability 99.95 percent monthly. Search and details endpoints 99.99 percent served from cache.

Booking confirmation success above 99.5 percent with defined playbooks. Error budget 0.5 percent. RPO zero for booking and payment event logs. RTO per region under 15 minutes with active active or warm standby. Supplier SLAs tracked and used for routing and throttling.

Security Privacy and Compliance

PCI DSS for payments with tokenization and no raw card storage. TLS and mTLS for service calls. Secrets in KMS with rotation. RBAC with least privilege. PII minimization and encryption at rest. GDPR and CCPA tooling for subject access and deletion. Data residency and retention controls per tenant. Bot and abuse protections for scraping and fake accounts. Audit logs for admin actions. DDoS and WAF rules at edge.

Multi Tenancy

Logical isolation by tenant id with row level security in Postgres. Per tenant domains and themes. Quotas and rate limits per tenant. Optional dedicated VPC and regional isolation for enterprise. Per tenant encryption keys and billing. Separate API keys and webhooks for partners.

APIs and Integrations

Public APIs

GET v1 search listings with location, dates, guests, filters, sort, cursor, limit.

GET v1 listings listingId details.

POST v1 bookings create with quoteld and payment token. Idempotency key required.

POST v1 bookings bookingId cancel or modify.

GET v1 bookings bookingId status.

Partner APIs

POST v1 partners ingestion ARI updates, GET v1 partners bookings, webhook for reservation confirmations.

Payments

Create payment intent, confirm with 3DS, capture, refund, payout to hosts. Webhooks for status updates. All endpoints versioned and idempotent.

CI CD and Deployment

Trunk based development with automated tests and security scans. Terraform for infrastructure with per environment workspaces. Progressive delivery with canaries on Search, Pricing, and Reservation, using shadow traffic for supplier adapters. Automatic rollback on SLO burn. Feature flags for pricing rules and filter experiments. Blue green deploys for stateful services with migration gates.

Observability and Operations

Tracing across gateway, services, and supplier adapters with OpenTelemetry. Metrics: search QPS, p95 and p99 latencies, cache hit rate, booking success, supplier error rates, payment auth success, hold expirations, SLO burn. Logs with PII redaction. Alerting on SLO burn, cache miss spikes, supplier timeouts, payment decline spikes, and hold leak detection. Dashboards per region, tenant, and supplier. Weekly ops review with action items. Runbooks linked from alerts.

Capacity Planning and Cost

Reference per region

Search cluster three data nodes and two masters. Redis cluster with twelve shards. Kafka three to five brokers with three day retention. Postgres with primary and two replicas, partitioned bookings table. Object storage for images and docs. CDN offloads media and JSON.

Estimated monthly cost from mid five to low six figures depending on region count, egress, and supplier mix. Cost levers include cache hit rate, price bucket depth, ARI polling frequency, and photo sizes.

Testing Strategy and Failure Drills

Unit and property tests for pricing and taxes. Contract tests for suppliers and payments. Load tests for search and booking spikes, chaos tests for supplier degradation, payment timeouts, and calendar conflicts. Replay tests from production like logs in staging. Security testing with SAST, DAST, dependency checks, and secret scanning. Disaster recovery drills including regional failover and data restoration from snapshots.

Phased Build Plan

MVP six to eight weeks

Core search with cached price buckets, listing details, simple reservation with prepay and email confirmation, single PSP integration, host portal basics, single region with multi AZ, metrics and dashboards.

Phase 2 two to three months

Multiple suppliers and PMS adapters, robust pricing and tax engine, cancellations and modifications, payouts to hosts, canary deploys, localization, fraud controls.

Phase 3 three to six months

Active active multi region, advanced personalization and map search, enterprise multi tenancy, data residency, rich ops tooling, and marketplace partner program.

Key Design Choices Trade Offs and Risks

Cache versus freshness

Edge caching reduces latency but risks stale pricing. Mitigate with short ttl, ETags, and background refresh.

Mark prices as from quote time and reprice at checkout.

Supplier heterogeneity

Different SLAs and semantics across suppliers complicate booking. Use a strict adapter contract, extensive contract tests, and circuit breakers.

Strong consistency for holds

Cap theorem forces tradeoffs; use scoped strong consistency for holds and booking commits while allowing eventual consistency for search.

Payment success and fraud

Stronger SCA reduces fraud but adds friction. Apply risk based SCA and retries with alternative routes.

Monitor auth and capture rates with fast failover PSPs.